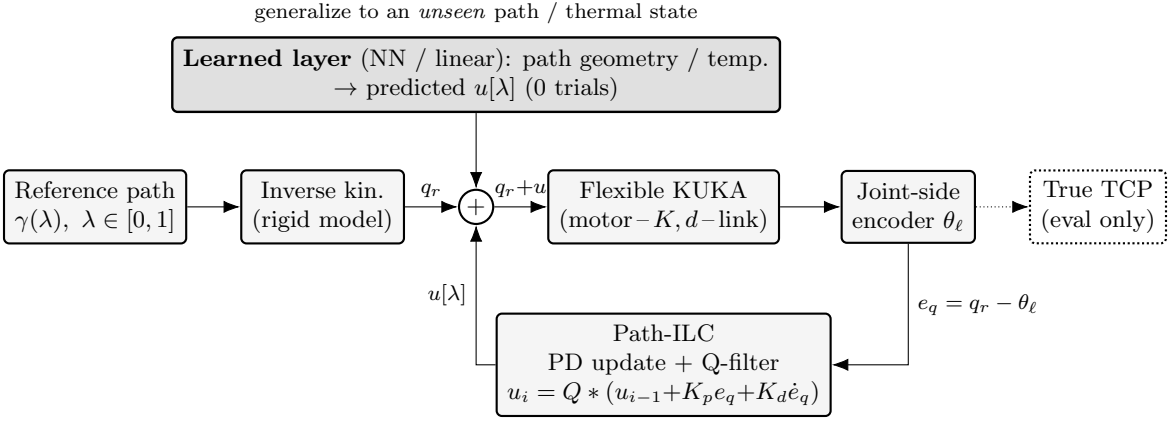

generalize to an *unseen* path / thermal state
Learned layer (NN / linear): path geometry / temp.
→ predicted $u[\lambda]$ (0 trials)
Reference path
$\gamma(\lambda),\ \lambda \in [0,1]$
Inverse kin.
(rigid model)
q_r
q_r+u
Flexible KUKA
(motor − K, d − link)
Joint-side
encoder θ_ℓ
True TCP
(eval only)
$u[\lambda]$
$e_q = q_r - \theta_\ell$
Path-ILC
PD update + Q-filter
$u_i = Q * (u_{i-1} + K_p e_q + K_d \dot{e}_q)$